Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

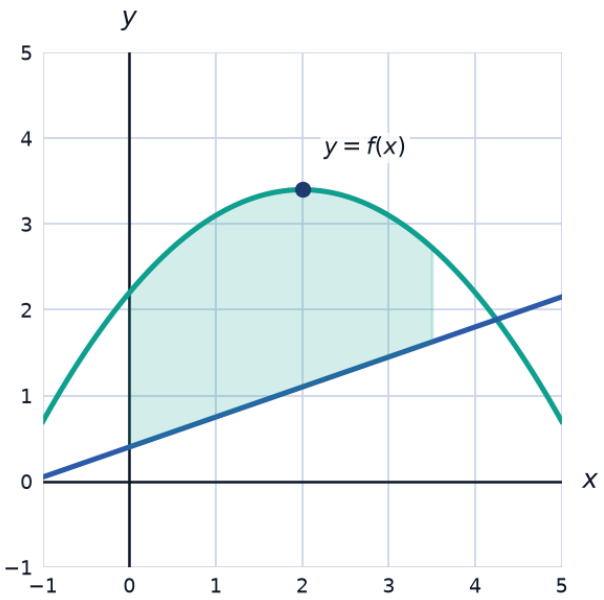
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Area and perimeter

A rectangle has the dimensions shown below. Find its area and perimeter.

2 [Foundational] • Target: Area and perimeter

Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

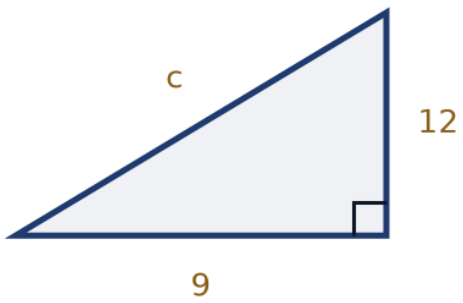


3 [Foundational] • Target: Area and perimeter

A triangle has base 15 and height 10. Find its area.

4 [Foundational] • Target: The Pythagorean theorem

A right triangle has the two legs shown below. Find the length of the hypotenuse c.



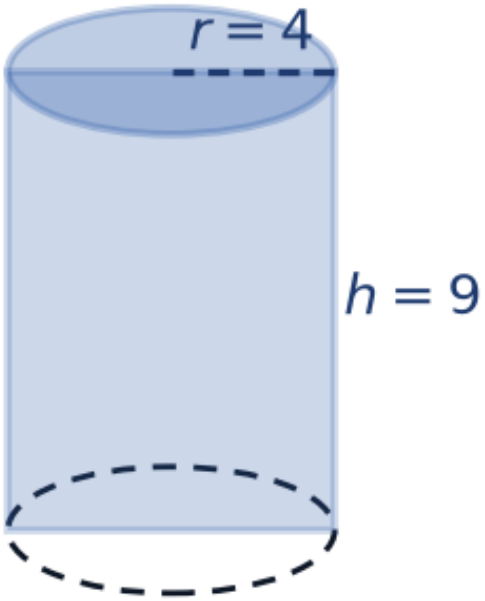
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5 [Foundational] • Target: The Pythagorean theorem

A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

6 [Foundational] • Target: Volume and surface area

Find the volume of a cylinder with radius 4 and height 9, in terms of π.



7 [Foundational] • Target: Volume and surface area

Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle ABC is similar to triangle DEF, with $AB = 8$, $DE = 12$, and $BC = 10$. Find EF.

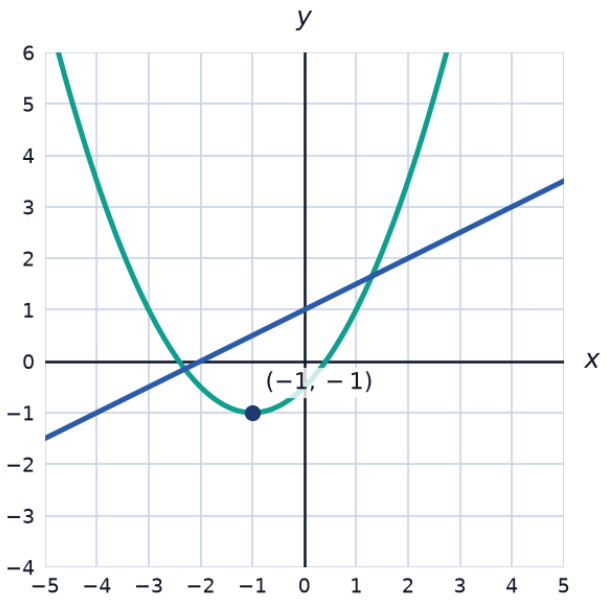
9 [Intermediate] • Target: Similar triangles and proportional reasoning

A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.

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10 [Intermediate] • Target: Angle relationships

Two parallel lines are cut by a transversal. If one angle measures 72°, find its corresponding angle and its co-interior (same-side interior) angle.



11 [Intermediate] • Target: Angle relationships

The three angles of a triangle are in the ratio $2:3:4$. Find each angle.

12 [Intermediate] • Target: Angle relationships

Two angles are complementary, and one is 34° more than the other. Find both angles.

13 [Intermediate] • Target: Coordinate geometry: distance and midpoint

Find the distance between the points $(3, -2)$ and $(9, 6)$.

14 [Intermediate] • Target: Coordinate geometry: distance and midpoint

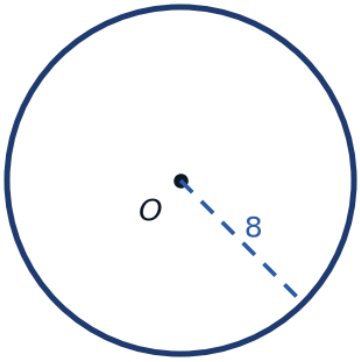
Find the midpoint of the segment connecting $(-5, 3)$ and $(7, -9)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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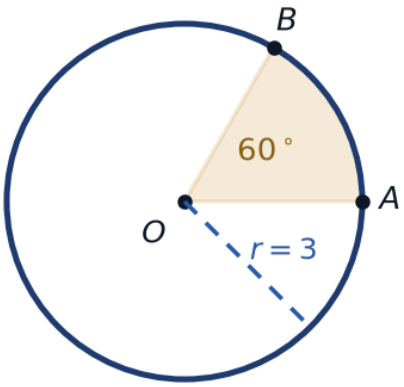
15 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A circle has the radius shown below. Find its area and circumference, in terms of π.



16 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 8.01 Exponentials & Logarithms

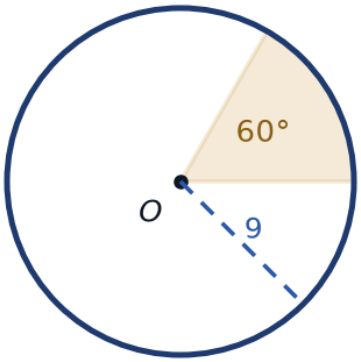
Write the equation of a circle with center $(3, -4)$ and radius 6.



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17 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 1.01 Heart of Algebra

A circle has the radius and shaded sector shown below. Find the area of the shaded 60° sector, in terms of π.



18 [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 2.01 Quadratics & Polynomials

A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π.

19 [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 3.01 Functions & Graphs

A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using $\pi \approx 3.14$, find the area of the garden not covered by the fountain, to the nearest whole number.

20 [Challenging] • Target: A well-designed word problem: designing a park • Cross-Lesson: 4.01 Ratios, Rates & Data

This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of π. (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of π.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Area and perimeter

A rectangle has length 23 and width 5. Find its area and perimeter.

a. $115, 56$

b. $56, 115$

c. $119, 56$

d. $115, 54$

22 [Foundational] • Target: The Pythagorean theorem

A rectangle has length 17 and width 15. Find its area and perimeter.

a. $255, 64$

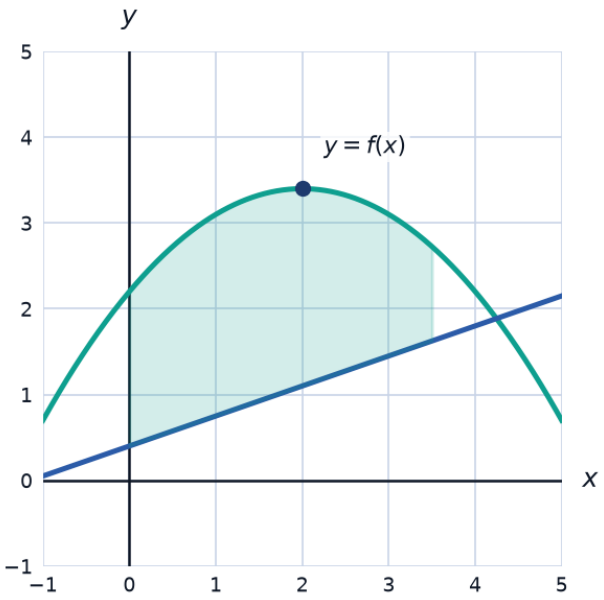
b. $64, 255$

c. $259, 64$

d. $255, 62$

23 [Foundational] • Target: Volume and surface area

A rectangle has length 27 and width 6. Find its area and perimeter.



a. $162, 66$

b. $66, 162$

c. $166, 66$

d. $162, 64$

24 [Foundational] • Target: Similar triangles and proportional reasoning

A rectangle has length 4 and width 13. Find its area and perimeter.

a. $52, 34$

b. $34, 52$

c. $52, 32$

d. $56, 34$

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25 [Foundational] • Target: Angle relationships

A rectangle has length 26 and width 16. Find its area and perimeter.

a. $416, 84$

b. $416, 82$

c. $420, 84$

d. $84, 416$

26 [Foundational] • Target: Coordinate geometry: distance and midpoint

A rectangle has length 20 and width 15. Find its area and perimeter.

a. $304, 70$

b. $70, 300$

c. $300, 68$

d. $300, 70$

27 [Foundational] • Target: Circles: area, circumference, and equation

A rectangle has length 13 and width 23. Find its area and perimeter.

a. $72, 299$

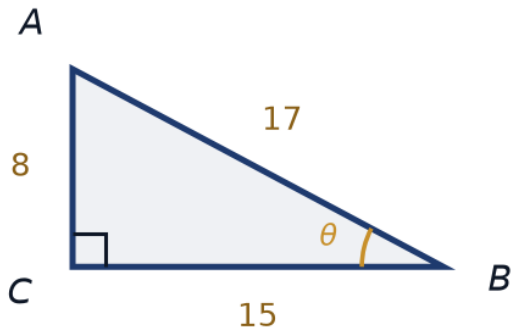
b. $303, 72$

c. $299, 70$

d. $299, 72$

28 [Foundational] • Target: Composite and shaded regions

A right triangle has legs 3 and 4. Find the length of the hypotenuse c.



a. 5

b. 7

c. 3

d. 9

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: designing a park

A right triangle has legs 7 and 24. Find the length of the hypotenuse c.

a. 27

b. 25

c. 31

d. 23

30 [Intermediate] • Target: Angle Relationships & Polygons

A right triangle has legs 20 and 21. Find the length of the hypotenuse c.

a. 27

b. 31

c. 41

d. 29

31 [Intermediate] • Target: Triangles, Similarity & Special Right Triangles

A right triangle has legs 11 and 60. Find the length of the hypotenuse c.

a. 71

b. 63

c. 61

d. 59

32 [Intermediate] • Target: Circles, Arcs & Tangents

A right triangle has legs 18 and 80. Find the length of the hypotenuse c.

a. 98

b. 84

c. 82

d. 80

33 [Intermediate] • Target: 3D Surface Area & Volume

A right triangle has legs 27 and 120. Find the length of the hypotenuse c.

a. 125

b. 123

c. 121

d. 147

34 [Intermediate] • Target: Area and perimeter

A right triangle has legs 9 and 12. Find the length of the hypotenuse c.

a. 13

b. 21

c. 15

d. 17

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35 [Intermediate] • Target: The Pythagorean theorem

Find the volume of a cylinder with radius 6 and height 4, in terms of π.

a. 576π

b. 24π

c. 150π

d. 144π

36 [Intermediate] • Target: Volume and surface area

Find the volume of a cylinder with radius 9 and height 7, in terms of π.

a. 2268π

b. 63π

c. 573π

d. 567π

37 [Intermediate] • Target: Similar triangles and proportional reasoning

Find the distance between the points $(-10, 2)$ and $(-15, -10)$.

a. 13

b. 11

c. 15

d. 17

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle relationships • Cross-Lesson: 2.01 Quadratics & Polynomials

Find the distance between the points $(-5, -9)$ and $(-14, 31)$.

a. 41

b. 43

c. 39

d. 49

39 [Challenging] • Target: Coordinate geometry: distance and midpoint • Cross-Lesson: 3.01 Functions & Graphs

A square with side length 19 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π.

a. $365 - 25\pi$

b. $361 - 25\pi$

c. $361 - 27\pi$

d. 361π

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40 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 4.01 Ratios, Rates & Data

A square with side length 16 has a circle of radius 4 removed from its center. Find the remaining area, in terms of π.

a. $260 - 16\pi$

b. 256π

c. $256 - 18\pi$

d. $256 - 16\pi$

41 [Challenging] • Target: Angle Relationships & Polygons • Cross-Lesson: 5.01 Statistics & Probability

Two angles are complementary. If the measure of one angle is 14° more than three times the other, what is the measure of the smaller angle?

a. 19°

b. 24°

c. 38°

d. 71°

42 [Challenging] • Target: Angle Relationships & Polygons • Cross-Lesson: 7.01 Trigonometry & Additional Topics

In ΔABC, the exterior angle at vertex C measures 125°. If one remote interior angle measures 55°, what is the measure of the other remote interior angle?

a. 55°

b. 65°

c. 70°

d. 75°

43 [Challenging] • Target: Triangles, Similarity & Special Right Triangles • Cross-Lesson: 8.01 Exponentials & Logarithms

Two similar triangles have perimeters of 24 cm and 36 cm. If the area of the smaller triangle is 32 cm^2, what is the area of the larger triangle?

a. 48 cm^2

b. 64 cm^2

c. 72 cm^2

d. 96 cm^2

44 [Challenging] • Target: Circles, Arcs & Tangents • Cross-Lesson: 1.01 Heart of Algebra

What is the area of a circle whose circumference is 16π?

a. 16π

b. 32π

c. 64π

d. 128π

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45 [Challenging] • Target: 3D Surface Area & Volume • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the total volume of a right circular cylinder with radius 5 cm and height 12 cm?

a. 60π cm^3

b. 120π cm^3

c. 300π cm^3

d. 600π cm^3

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